

INTRODUCTION

- Memory is a vital function that helps us process information, make decisions, and learn effectively. It is a primary function we use everyday, and being able to understand and enhance it can reap massive benefits.
- Memory consists of three main steps: encoding, retention, retrieval. Each of these tasks are distinct from each other and are essential for processing and storing information in the brain. Consequently, any malfunction in these processes can result in difficulty performing simple tasks and induce stress and frustration in individuals.
- Metacognitive awareness is the ability to identify when the brain is focused and which areas perform the best during specific tasks. This is essential in forming memories as it allows individuals to recognize how they perform best, and utilize those strengths to their benefit. Hence, it can also help identify degradation in memory, and achieve early intervention to prevent the progression of any memory problems.
- The Muse 2 is an EEG headset with advanced technology, making it an excellent tool to achieve metacognitive awareness for an affordable price.
- In this project, I used the Muse 2 headset to create an accessible and affordable program which helps individuals achieve metacognitive awareness.

PROBLEM

Around the world, over 55 million individuals suffer from dementia, with 60-70% of that population experiencing Alzheimer's. Around a quarter of the population on the planet experience memory problems such as brain fog and other types of forgetfulness. These types of problems can cause difficulty in performing basic tasks, and can lead to frustration and stress. Most of the time, people cannot identify and get these problems diagnosed because of accessibility struggles and issues with EEG procedures in the lab. These reasons are listed below:

- The cost for lab EEGs can exceed \$5,000 per day for certain tests, which is one reason people might feel that a few forgetful instances is not worth getting checked, as the majority of people do not have access to this kind of money.
- The wait times for EEG tests are extremely long, as many many hospitals do not have access to many EEG systems. Even after the test is done, it can take weeks for the participant to receive their results, which can lead to any underlying conditions worsening over that periods of time.
- During the test, many participants with thick, curly, or braided hair might be asked to cut or remove their hair, which is a main reason why many people might not want to participate in a EEG test, as some people naturally have hair that might not be able to work with an EEG.
- Hospitals and academic institutions which have EEGs might be far away or inaccessible for many people, especially those living in poor or rural areas. The process for applying and receiving EEG tests is long and complicated, taking a long time for application and selection. This might discourage many to not take these tests because of the inaccessibility.
- The test itself could take many hours and can be tedious and uncomfortable for many, and the stress of a lab environment might affect the results from the test, potentially leading to false results and misdiagnosis and treatment, which can worsen the problem, or cause healthy individuals to receive incorrect and possibly dangerous treatment.

My project aims to fix these problems and provide a cheap, fast, and low pressure test which can give participants accurate results and highlight their strengths and weaknesses in memory. It will make memory testing much more accessible and efficient to people who have low income and live in rural or poor areas.

It will also be able to provides strengths which individuals can utilize to further expand their memory and use techniques which will help them succeed academically and beyond in life.

LIMITATIONS

- The first limitation is that the Muse 2 headset does not possess the same capability as a proper EEG cap, or EEGs used in labs. Though it has very reliable data, the limitation of the area of the electrodes makes some of the brainwaves not entirely accurate, as the headset is initially designed for relaxation.
- Also, though there is filtration, some signals from eyes and muscle movements can affect the data, as well as oil and hair between the electrodes can weaken the connection.
- The Bluetooth connection between the headset and the laptop may falter at times, providing incorrect time for some of the events.
- The participant may also have external factors influencing them, potentially determining how they do on the game, and whether the brainwaves that are recorded are accurate to how their brain works.
- The project also does not measure long-term memory, which is very significant, and the program does not test for long periods of time.

IMPROVEMENTS

- With the help of the MuseLSL (Lab Streaming Layer), this program can be made to stream more advanced and accurate data from the EEG headset (next level in my project).
- To minimize the interference of external factors, a baseline test could be added to the beginning of the game so that the test will be accurate no matter who or when the test is taken.
- The program could be enhanced to give a deeper analysis, and potentially try and visual live data on a graph. A validation test can be added to affirm the accuracy of the program with an advanced institutional involvement (ex. University of Calgary).

FUTURE APPLICATIONS

- This project allows individuals to achieve metacognitive awareness using an accessible program and affordable equipment, which delivers accurate results efficiently. The ability to identify one's cognitive strengths and weaknesses can have immense potential in home and clinical settings.
- In older individuals, it can play a significant role in regularly assessing their memory and tracking any potential memory disorder in an accessible and affordable way.
- In younger groups, it can be used as a tool to help support cognitive assessments, such as IQ or ADHD tests, showing the participants' true level of engagement and mental stimulation during the test.
- The program can be used as a mobile or desktop app which is easily accessible from home with support for the Muse 2 headset.

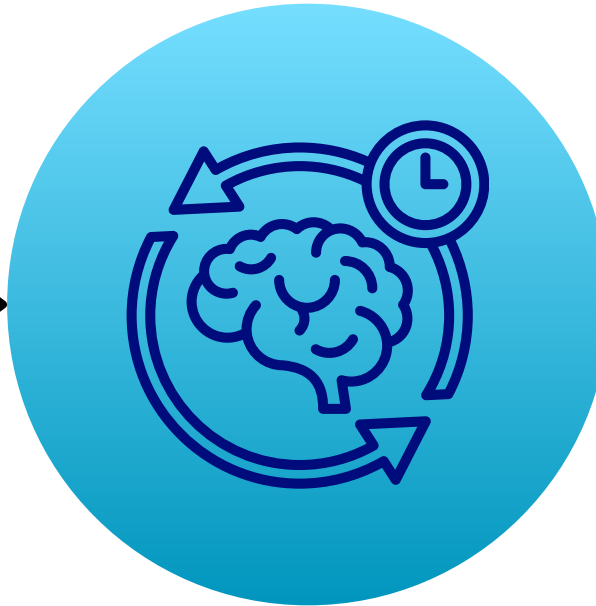
MUSE 2 HEADSET

MEMORY GAMES

RECORD DATA



The Muse 2 headset is used to record the brainwaves of the participants. It is equipped with an advanced EEG system and is available for an affordable price.



All 4 memory games are played in Unity, and timestamps for the beginning and ending of the games are recorded and used for analysis.



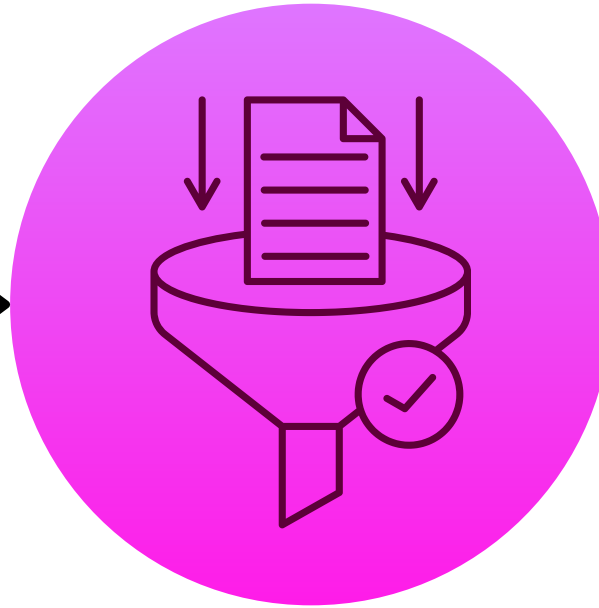
While the person is playing the games, all the data is streamed from the headset to the laptop using Brainflow.

SAVE DATA



The data is saved as FIF and CSV files which are ideal for processing and filtering the data using tools in Python.

FILTRATION



The data is filtered through man different tools in order to eliminate any artifacts from eyes and muscles.

PRESENTATION



Once the data is processed, it is organized into a neat report which highlights the person's results and achieves metacognitive awareness.

FILTRATION

- The filtration process consists of many steps to ensure that the data is accurate and removes any unnecessary components from the files. It uses mne-python to access EEG filters.
- The first step is to apply a bandpass filter to the data. This filter is set for 1-40 Hz, and this ensures that the data is purely the participants brainwaves, as the highest brainwaves only goes up to 40 Hz. It removes extra waves from muscles and eyes,
- The second step is to compute an average reference from all of the brainwaves, as the natural setting can be to reference all the data from the two frontal sensors. This makes sure that all the data is computed equally from all four electrodes, retaining accuracy for the report.
- The third step is to use ICA (Independent Component Analysis) to further filter out artifacts such as eye and muscles movement more precisely and accurately. This analyzes the data and finds more specific instances of eye and muscle artifacts and removes them from the data. This ensures accuracy in the results.
- The fourth step would be compute the average bandpower of each wave using the Welch method to identify the strength of each brainwave during each task. In the Welch's method, the signals are overlapped over each other, windowed, and then a periodogram is extracted for each of them, which in them averaged to produce a single smooth average for the data.
- The last step is to extract the engagement and focus use the theta and beta brainwaves and using them to generate the final report. They are organized to determine in which games the participant was most engaged and focused in, and achieve metacognitive awareness.

RECORDING & SAVING

- In order to stream data from the Muse 2 headset to the laptop, I had to use a third-party software called Brainflow. This software uses a BLE (Bluetooth Low Energy) system to connect with the headset.
- Brainflow will take data from the 4 electrodes on the headset and stream them to the laptop so that the data is ordered properly and accurately received from all 4 electrodes.
- While the user is playing the memory games, Unity is sending the Python app timestamps for the beginning and ending of each game. The timestamps are sent through UDP (User Datagram Protocol) markers, which are extremely precise and ultra fast, ensuring that the data is transmitted accurately.
- After, the EEG data is organized into a FIF file, ensuring all the data from the four electrodes is retained. The UDP timestamps are organized into a CSV file, which are then used with the FIF files to ensure that the data is orientated properly with the beginning and end of the games.
- The data is now saved and ready to be processed and filtered to generate the report.